ClipSAT : IB Diploma Mathematics: Analysis & Approaches / Applications

Unit 3.01: Topic 3: Geometry, Trigonometry & Voronoi Diagrams

Name: _____

Q1. In the diagram, triangle ABC is shown with coordinates A1, 1, B3, 1, and C3, 5. What is the exact length of the side AC in simplest radical form?

[ ] A: 4
[ ] B: 5
[ ] C: 2sqrt{5}
[ ] D: 4sqrt{2}

Q2. In the given triangle ABC, AB = 10, AC = 15, and BD = 6. D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.

[ ] A: 9
[ ] B: 10
[ ] C: 15
[ ] D: 22.5

Q3. Volume of cone with radius 3, height 4:

[ ] A: 12pi
[ ] B: 16pi
[ ] C: 36pi
[ ] D: 48pi

Q4. In the given right-angled triangle ABC, angle B is 90 degrees. Point D is on BC such that BD = 4 and DC = 9. If AD is the altitude to the hypotenuse AC, find the length of AD.

[ ] A: 6
[ ] B: 8
[ ] C: 12
[ ] D: 15

Q5. In the triangle ABC, AB = 10, AC = 15, BD = 6, and D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.

[ ] A: 7
[ ] B: 9
[ ] C: 10
[ ] D: 12

Q6. In the diagram, triangle ABC has angle ABC equal to 90 degrees. Point D lies on AC such that BD is perpendicular to AC. If AD = 4 cm and DC = 9 cm, find the length of BD.

[ ] A: 5 cm
[ ] B: 6 cm
[ ] C: 7 cm
[ ] D: 3 cm

Q7. In the right-angled triangle ABC shown below, the length of side AB is 10 units and the length of side BC is 6 units. The angle ACB is 90 degrees. Find the length of side AC.

[ ] A: 10
[ ] B: 6
[ ] C: 8
[ ] D: 16

Q8. In the right-angled triangle ABC, shown below, AD is the altitude to the hypotenuse BC. If BD = 4 and DC = 9, find the length of the altitude AD.

[ ] A: 5

[ ] B: 6

[ ] C: 7

[ ] D: 8